Paper A v0.1 — The Physical (SU(3)) Yang–Mills Mass Gap via D_IV^5 Spectral Geometry. Companion to Paper B (substrate uniqueness). The wall map collapsed (Sunday 2026-06-21 PM): W4 dissolved (Paper B), W3 folded (HS net-compat via BGL), W1 FOLDED onto W2 (the D_IV^5 spectral theory already supplies Osterwalder–Schrader data — rigorous Hilbert space H^2, gapped Casimir Hamiltonian, Szegő reflection positivity, 4D conformal/Poincaré, clustering — so the famous from-scratch-on-R^4 construction is sidestepped). The ENTIRE prize reduces to ONE question: is the OS-reconstructed gapped 4D theory genuinely INTERACTING Yang–Mills, or a rigorous-but-trivial generalized-free gapped theory? That question = W2 (the cross-channel glueball spectrum match) = #418 (is the gauge algebra genuinely non-abelian su(3)?), three faces of one gauge-structure question, all in so(7) ⊃ g_2 ⊃ su(3). Honest tiers: Δ = C_2 = 6 SOLID; OS data reduction SOLID-modulo-bounded-checks; interacting-vs-free OPEN = W2; the result is a TWO-CONDITION CONDITIONAL ('given the cross-channel spectrum closes AND the OS bounded-checks hold, physical SU(3) YM has gap C_2 = 6'), explicitly NOT a claim that YM is proved.

Lyra (Claude Opus 4.8) — Casey Koons, PI; Grace (OS reduction + net-compat + W2 build), Elie (#418 Sc

2026-06-21 Sunday

The Physical Yang–Mills Mass Gap via D_IV^5 Spectral Geometry

Paper A of the two-paper package (companion: Paper B, substrate uniqueness)

Abstract

We study the Yang–Mills mass gap for the physical gauge group SU(3) — color — derived (not assumed) from the substrate D_IV^5 in the companion Paper B. We show that the mass gap value is the exact integer $\Delta = C\_2 = 6$, the lowest nontrivial Casimir eigenvalue of the compact dual $Q^5 = SO(7)/[SO(5)\times SO(2)]$. The construction does not attempt the famous open problem of building 4D Yang–Mills from scratch on flat R^4; instead, the D_IV^5 spectral theory directly supplies Osterwalder–Schrader reconstruction data — a rigorous Hilbert space, a gapped Hamiltonian, reflection positivity, 4D conformal/Poincaré covariance, and clustering — so the existence of a rigorous gapped 4D theory is obtained by classical harmonic analysis on a bounded symmetric domain. The entire problem then reduces to a single question: is this rigorous gapped theory genuinely interacting Yang–Mills, or a (also-axiom-satisfying, also-gapped) generalized-free theory? We show this question is identical to the cross-channel glueball spectrum match (W2) and to whether the gauge algebra is genuinely non-abelian su(3) (the bulk-color realization). We state the result honestly as a two-condition conditional and present the spectrum test as the decisive, falsifiable probe.

1. Scope: the physical theory, not "all G"

The Clay problem asks for a mass gap for *every* compact simple gauge group. BST addresses the physical theory: G = SU(3), color. The companion Paper B proves D_IV^5 is the unique substrate and that SU(3) is derived, not chosen (criteria-innocent uniqueness; $N\_c = 3$ forced dimension-free, Section 7). So "all G" is not BST's claim and not, on the BST view, a physical question: there is one substrate and it yields one gauge group. This paper makes claims only about the physical theory and is explicit about that scope (Section 9).

2. The mass gap value: $\Delta = C\_2 = 6$ (SOLID)

The Yang–Mills Hamiltonian is identified with the D_IV^5 Casimir, whose spectrum on the compact dual Q^5 is the scalar-Laplacian tower $\{k(k+5)\}\_\{k\geq 0\} = \{0, 6, 14, 24, \ldots\}$. The lowest nontrivial eigenvalue — the gap — is

$\Delta = C\_2 = 6$,

the SO(7) vector Casimir. This is an exact integer from representation theory, not a numerical estimate. Physical anchoring: proton $= C\_2\cdot\pi^5\cdot m\_e = 938.25$ MeV; the 0^{++} glueball $= c\_2\cdot\pi^5\cdot m\_e = 1720$ MeV ($c\_2 = 11$, the 2-form Hodge gap). Tier: SOLID (rep theory + the substrate Hamiltonian identification).

3. W4 dissolved: the gauge group is derived (Paper B)

A referee objection — "you treat only SU(3)" — is answered not by proving all G but by showing G is not a free input. Paper B establishes D_IV^5 as the unique irreducible Hermitian symmetric domain meeting prior, dimension-innocent criteria, forcing $\dim\_C = 5$ and $N\_c = 3$ (the multiplicity bound is fixed by the rank via the proved Wallach Bottleneck Theorem T1829; the dimension enters only through $m\_s = n-2$). So SU(3) is substrate-derived. W4 (the "all G" wall) is dissolved.

4. W1 folded: the D_IV^5 spectral theory supplies the OS data (the key reframe)

The hard part of the Clay problem — *constructing* a rigorous interacting 4D QFT — is famous because building it from scratch on flat R^4 (path integral, axioms) is open. BST does not do that. The Osterwalder–Schrader reconstruction theorem builds a QFT from a data set, and the D_IV^5 spectral theory already carries that data:

OS datum	D_IV^5 realization	tier
rigorous Hilbert space	the Hardy space $H^2(D\_IV^5)$	SOLID
Hamiltonian with a gap	the Casimir, gap $= C\_2 = 6$	SOLID (Section 2)

OS datum	D_IV^5 realization	tier
reflection positivity	engaged (F270): D_IV^5 is tube-type (Cayley → $T(\Omega)$ over the Lorentz cone); Euclidean-time reflection Θ = the cone involution; OS reflection positivity (free level) = the Cauchy-Szegő reproducing-kernel positivity, automatic by RKHS	SOLID (free level)
4D conformal/Poincaré covariance	$SO(4,2) \subset SO(5,2)$, UV-conformal/IR-gapped (= asymptotic freedom); Szegő-equivariance	SOLID-structural
clustering / unique vacuum	follows from the spectral gap Δ = C_2 = $6 > 0$ (exponential clustering)	SOLID

So the existence of a rigorous gapped 4D theory comes from classical harmonic analysis on a bounded symmetric domain — the open from-scratch construction is sidestepped, not solved. (Grace's reduction.) The OS bounded-checks are now engaged, not labeled (F270): at the free/spectral level reflection positivity (Szegő RKHS-positivity on the tube, Θ = cone involution), conformal covariance, clustering (from the gap), and temperedness all hold; the single non-free-level piece is the interacting upgrade of reflection positivity (RP for the full interacting measure / higher correlators), which is W2 itself (is the theory genuinely interacting?). Honest note (Clay framing): the OS axioms are not the route to the *gap* — spectral geometry gives that directly — they are what makes the W1 fold rigorous ("a gapped 4D theory exists" as an OS-reconstructible object). Tier: SOLID at free level; interacting-RP = W2.

5. W3 folded: net-compatibility via Bisognano–Wichmann (SOLID-CONDITIONAL)

The HS isometry intertwines the bulk and boundary operator nets, not merely the Hilbert spaces: HS is SO(5,2)-equivariant (Hua–Korányi), and both the bulk Rehren net and the boundary net are modular reconstructions of the *same* SO(5,2) positive-energy representation via Bisognano–Wichmann (Brunetti–Guido–Longo). So locality/causality transfer across HS. Tier: SOLID-CONDITIONAL on the BGL hypothesis-checks (Grace's net-compatibility lemma, being brought to SOLID in parallel).

6. The one remaining question: interacting, or generalized-free?

After Sections 3–5, the entire prize is a single question:

Is the OS-reconstructed gapped 4D theory genuinely interacting Yang–Mills, or a generalized-free gapped theory? (A generalized-free field also satisfies the OS axioms and also has a gap — so the axioms alone do not decide it.)

A trap we explicitly avoid. The boundary observables form a non-commutative Toeplitz operator algebra on H^2 (Hankel-compact commutators). It is tempting to conclude "non-commutative □ interacting." This is false: a *free* field also has a non-commutative operator algebra (the CCR). Operator non-commutativity is not interaction; interaction means nonzero connected n-point functions ($n \geq 3$).

The real locus of the interaction. The Yang–Mills self-interaction is the vertex $A \wedge A = [A,A]$, nonzero precisely because the gauge algebra is non-abelian. Therefore:

interacting (not generalized-free) □ the gauge algebra is genuinely non-abelian su(3) □ the cross-channel glueball spectrum matches SU(3) (W2).

Three faces of one gauge-structure question, all living in the $so(7) \supset g_2 \supset su(3)$ of Section 8. Two pieces of evidence toward the interacting side:

- The bulk-color algebra closes as non-abelian su(3) (the bilinear realization). The linear Toeplitz operators would generate a Heisenberg/oscillator algebra (not simple); the bilinear (Schwinger) realization closes into su(3) — verified explicitly (CCR + all 81 gl(3) commutators + the A_2 structure relations, Elie toy 4301). The remaining piece is the substrate identification (that the bulk-color Toeplitz operators are these bilinears) — framework, in progress.
- The Casimir tower is non-additive (a lead, not banked). The tower {0, 6, 14, 24, …} has its level-2 entry at 14, not 2×6 = 12; a free multi-particle spectrum would be additive, and non-additivity is the fingerprint of binding/interaction. *Caveat:* this argues interaction only insofar as the relevant levels are multi-particle rather than single-particle tower states; the cross-channel computation (Section 7) is what sharpens it into evidence.

7. The decisive probe: the cross-channel glueball spectrum (W2) — NAMED-OPEN, structural-tier

The cross-channel glueball spectrum is the decisive, falsifiable probe of whether the OS-reconstructed theory is genuinely interacting SU(3) Yang–Mills (Section 6). Its honest status in v0.1 is named-open at structural tier: the mass gap value C_2 = 6 (Section 2) and the 0^{++} anchor (c_2 = 11 → 1720 MeV) are SOLID and independent of it — they do not depend on the full cross-channel match. What is open is the *full spectrum* across all J^PC channels.

A blind forward-prediction test is set up (and is the work in progress, not banked into v0.1): - Pre-registration (from operator structure only, no lattice number): the glueball channels are assigned by standard operator content — 0^{++} = Tr(F^2), 0^{-+} = Tr(F $\tilde{F}$), 2^{++} = the conserved stress tensor (protected dimension $\Delta = 4$, $\gamma = 0$), 1^{+-} = Tr(F[D,F]) (canonically heaviest); oddballs (0^{+-}, 1^{-+}, 2^{+-}) are forbidden at two gluons and are the clean, unmixable experimental channels. - The verdict reduces to a single operator quantity. With 2^{++} pinned by conservation and 1^{+-} separated by canonical dimension, the only degeneracy to lift is 0^{++} vs 0^{-+}, which are identical in every representation label and differ only by parity (Hodge-*). Their split is the □-graded spectral asymmetry Tr(□ $\hat{H}$) of the bulk Casimir on the 2-form sector — an index/η-invariant, the parity-odd (Pontryagin / topological, Witten–Veneziano-class) part of the operator. - A falsifier built into the geometry: D_IV^5 is Kähler ((1,1) curvature); if the □-graded asymmetry vanishes, 0^{++} and 0^{-+} are predicted* degenerate* — a clean miss against the lattice (which splits them ~1.5×); if nonzero, that asymmetry is the predicted split.

Honest disposition: the spectrum-matches-to-completeness condition of Section 9 is not yet established; it is a *named-open* computation (the blind forward-prediction test) that will confirm, honestly bound, or falsify the interacting reading. The decay-side mechanism is mixing (glueball ↔ q$\bar{q}$ co-assembly overlap on the shared Hardy space — matching the f0(1500)/f0(1710) phenomenology), not a "dump." The operator-class taxonomy and the factorization of Section 6 are supporting structure for this open test, not banked predictions.

8. The so(7) unification (LEAD-STRENGTHENED)

Color and the Yang–Mills spectrum live in one algebra. su(3) ⊂ g_2 ⊂ so(7), where g_2 is the octonion 3-form stabilizer (14 = 8 ⊕ 3 ⊕ $\bar{3}$ under su(3); the 7 = 3 ⊕ $\bar{3}$ ⊕ 1, the 1 the color singlet), and so(7) is the compact dual isometry of Q^5 — the *same* so(7) whose Casimir spectrum is the glueball tower of Section 2. So the color group and the gauge-boson spectrum are unified in one so(7); g = 7 is the so(7) vector = g_2 fundamental. Note the duality: color is not a geometric isometry of the noncompact domain D_IV^5 (su(3) □ so(5,2) compactly — three rep-dimension obstructions), but is a geometric subalgebra of the compact-dual isometry so(7); on the domain side it is the operator algebra on H^2. Tier: LEAD-STRENGTHENED — the algebraic picture is whole; the operator realization on H^2 (the bilinear-Schwinger identification) is in progress.

9. Honest scope: the two-condition conditional

We do not claim to have proved the Yang–Mills mass gap. The honest statement is:

Given (i) the OS bounded-checks (reflection positivity, locality via Section 5) hold, and (ii) the cross-channel spectrum closes to completeness (Section 7, identifying the theory as interacting SU(3) YM rather than generalized-free), the physical SU(3) Yang–Mills theory exists as a rigorous 4D QFT with mass gap Δ = C_2 = 6.

What is SOLID independent of the conditions: the gap *value* C_2 = 6 (Section 2); the substrate-derivation of SU(3) (Paper B); the OS *data* (Section 4); the so(7) unification at the algebraic level (Section 8). What is conditional/open: that the OS-reconstructed theory is genuinely interacting (Sections 6–7) — the one real question. This is a sharpening of the Clay problem to a single decidable test, not a claim of its solution.

10. Open items and falsifiers

- W2 cross-channel verdict (Section 7): the decisive probe; confirms / bounds / falsifies. Falsifiable — the blind targets can miss.
- #418 substrate identification: that the bulk-color Toeplitz octet *is* the bilinear-Schwinger su(3) on H^2 (the algebra closure is verified; the identification is in progress).
- OS bounded-checks (Section 4): reflection positivity and locality to executed-rigor (Section 5 net-compat to SOLID).
- Falsifier: if the cross-channel spectrum does not match SU(3) ratios, the theory is generalized-free and the "interacting YM" reading fails — the gap value C_2 = 6 and Paper B survive, but Paper A's interacting claim does not.

Count HOLDS 4 of 26. SU(3) scope. Paper A v0.1: the wall map collapsed to one question (interacting vs generalized-free = W2 = #418), with the gap value SOLID and the result an honest two-condition conditional. §7 integrates the W2 verdict when it lands. INTERNAL.

Draft v0.1. Spine: Grace's OS reduction (W1 folded) + net-compat (W3) + Lyra's W1=#418=W2 reduction (F265) + Elie's Schwinger closure (4301) + the so(7) unification + Paper B (W4). Honest two-condition-conditional framing per the standing scope discipline. Companion to Paper B v0.5.

— Lyra, Sun 2026-06-21 (date-verified). Paper A v0.1, wall-map-collapsed.